

INPUT (generator-agnostic)

Final-state particles (any generator)
 $\{(\text{pdg}_i, p_{x,i}, p_{y,i}, p_{z,i})\}$, vertex (x, y, z) , nucleus (Z, A)

Input pipeline
 drop ν , n , pseudo-particles, remnants; hadron KE ≥ 50 MeV
 pdg $\rightarrow$ 11 species classes; π^0 kept as one token

Tokens $n \leq 32$ particles
 class embedding $\oplus$ MLP($\text{asinh } \vec{p}, \log |p|, \hat{p}$) $\rightarrow$ 128
global token = MLP(x, y, z, Z, A) $\rightarrow$ 128

ENCODER

Set encoder
 transformer, 4 layers, $d=128$, 4 heads, pre-LN
 no positional encoding, key-padding mask
 $\Rightarrow$ event embedding $z \in \mathbb{R}^{128}$ (global token)
 $\Rightarrow$ token states $h_i \in \mathbb{R}^{128}$

HEADS: sampled in this order $\downarrow$

Event classification heads (MLP on z)
 $p(\text{reco muon candidate} \mid z)$ Bernoulli
 $p(\text{MINOS matched} \mid z)$ Bernoulli
 $p(\text{charge} < 0 \mid z)$ Bernoulli
 $p(N_{\text{prongs}} \mid z), N \in \{0..8\}$ categorical
 flags, N

Vertex-plane head (MLP)
 in: c_1 , phase of true z in the 22 mm plane cycle
 out: 9 classes: unsnapped / plane $\delta \in [-3, 3]$ / far plane
 flags, N

Event flow (conditional flow matching, 9-d)
 $x_1 = \text{muon } (\log P/P_{\text{true}}, \Delta\theta_x, \Delta\theta_y)$,
 vertex residual $(\Delta x, \Delta y, \Delta z)$,
 $\log E$: recoil, non-vertex 100 mm, isolated blobs
 velocity MLP 5×768 ; cond. $c_1 = [z, \text{flags}, \text{emb}(N)]$
 flags, N, x_1

Prong set flow (masked flow matching, $N \times 10$)
 per prong: has-kinematics, $\hat{u}$ (3), $\log P_\pi$, proton fit,
 $\log P_p$, logit score, exiting, primary
 equivariant transformer, 3 layers: self-attn over prongs,
 cross-attn to h_i ; cond. $[c_1, x_1]$; flags dequantised

OUTPUT

Decoder
 inverse transforms (asinh, log, ratio tables)
 vertex z : snap to plane if class > 0
 prong table $\rightarrow$ pion/proton/sec-proton blocks
 beam-frame angles, on-shell E , fill values

Pruned MasterAnaDev ntuple
 80 branches, same names / types / _sz
 as the open-data tuple

solid = data flow; dashed = conditioning on sampled quantities; dotted = cross-attention to token states.
 Sampling order: classification heads $\rightarrow$ vertex class $\rightarrow$ event flow $\rightarrow$ prong set flow $\rightarrow$ decoder. Training: joint, all losses, warm start of the prong flow from the event-level model.